

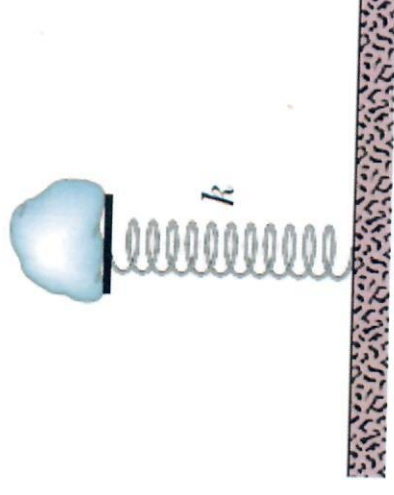
The figure shows an 8.00 kg stone at rest on a spring. The spring is compressed 10.0 cm by the stone.

- (a) What is the spring constant?
- (b) If the stone is pushed down an additional 30.0 cm and then released, what is the elastic potential energy of the compressed spring just before that release?
- (c) What is the change in the gravitational potential energy of the stone–Earth system when the stone moves from the release point to its maximum height?
- (d) What is that maximum height, measured from the release point?

Die figuur dui 'n 8.00 kg klip wat op 'n veer rus. Die veer word saamgepers deur 'n afstand van 10.0 cm.

- (a) Wat is die veerkonstante?*
- (b) Die klip word 'n verdere 30.0 cm afgedruk en losgelaat. Wat is die elastiese potensieele energie van die saamgepersde veer net voor dit losgelaat word?*
- (c) Wat is die verandering in die gravitasie-potensieele energie van die klip-Aarde stelsel wanneer die klip van die loslaat punt beweeg na die maksimum hoogte.*
- (d) Wat is die maksimum hoogte, gemeet van die loslaat punt?*

(10)



Tutorial Test 7-1

$$m = 8.00 \text{ kg}$$

$$\Delta y_{\text{spring},1} = -10.0 \times 10^{-2} \text{ m}$$

$$\Delta y_{\text{spring},2} = -40.0 \times 10^{-2} \text{ m}$$

$$a) \Sigma F_y = ma_y = 0 = F_s - mg \checkmark$$

$$F_s = -ky = mg$$

$$k = -\frac{mg}{y_{\text{spring},1}} \checkmark = \frac{-8.00 \text{ kg}(9.81 \text{ m/s}^2)}{-10.0 \times 10^{-2} \text{ m}}$$

$$= 784 \text{ N/m} \checkmark$$

$$b) U_s = \frac{1}{2} k y^2$$

$$= \frac{1}{2} (784 \text{ N/m}) (\underbrace{-40.0 \times 10^{-2} \text{ m}}_{\text{SI units}})^2 \checkmark$$

$$= 62.7 \text{ J} \checkmark$$

$$c) \text{ At max height } U_{sf} = 0, U_{gf} = \text{max}, \text{ let } U_{gi} = 0 \text{ at release point} \checkmark$$

$$K_i = K_f = 0$$

$$\Delta E_{\text{mech}} = \Delta K + \Delta U_g + \Delta U_s = 0 \checkmark$$

$$\Delta K = 0 \therefore \Delta U_g = -\Delta U_s = -(0 - 62.7 \text{ J}) = 62.7 \text{ J} \checkmark$$

$$d) \text{ From } E_{\text{mech}} = \Delta K + \Delta U_g + \Delta U_s = 0 \checkmark$$

$$K_i + U_{gi} + U_{si} = K_f + U_{gf} + U_{sf} \checkmark, K_i \text{ and } K_f = 0.$$

$$0 + 0 + \frac{1}{2} k y_i^2 = 0 + m g y_f + 0$$

$$y_f = \frac{\frac{1}{2} k y_i^2}{mg} \checkmark = \frac{\frac{1}{2} (784 \text{ N/m}) (-40.0 \times 10^{-2} \text{ m})^2}{(8.00 \text{ kg})(9.81 \text{ m/s}^2)}$$

$$y_f = 0.80 \text{ m} \checkmark$$

A horizontal force of magnitude 35.0 N pushes a block of mass 4.00 kg across a floor where the coefficient of kinetic friction is 0.600.

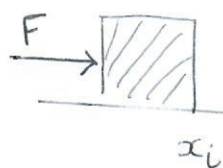
- (a) How much work is done by that applied force on the block–floor system when the block slides through a displacement of 3.00 m across the floor?
- (b) During that displacement, the thermal energy of the block increases by 40.0 J. What is the increase in thermal energy of the floor?
- (c) What is the increase in the kinetic energy of the block?

’n Horisontale krag van grootte 35.0 N stoot ’n blok van massa 4.00 kg oor die vloer waar die kinetiese-wrywingskoeffisient 0.600 is.

- (a) Hoe veel arbeid word verrig deur die toegepaste krag op die blok-vloer stelsel wanneer die blok deur ’n verplasing van 3.00 m oor die vloer gly?*
- (b) Tydens die verplasing, styg die termiese energie van die blok met 40.0 J. Wat is die toename van die termiese energie van die vloer?*
- (c) Wat is die toename in die kinetiese energie van die blok?*

(10)

Tutorial Test 7-2



$$m = 4.00 \text{ kg}$$

$$F = 35.0 \text{ N}$$

$$\mu_k = 0.600$$

$$\Delta x = 3.00 \text{ m}$$

$$\begin{aligned} \text{a) } W_F &= F \Delta x \cos \theta \\ &= 35.0 \text{ N} (3.00 \text{ m}) \cos 0^\circ \checkmark \\ &= 105 \text{ J} \checkmark \end{aligned}$$

$$\text{b) } \Delta E_{\text{int}} = f_k \Delta x = \mu_k N \Delta x$$

$$\sum F_{\text{block}, y} = m a_y = 0 = N - mg \quad \left. \begin{array}{l} \checkmark \\ \therefore N = mg \end{array} \right\} \text{ must have Newton's 2nd Law}$$

$$\begin{aligned} \therefore \Delta E_{\text{int}} &= \mu_k mg \Delta x \checkmark = (0.600)(4.00 \text{ kg})(9.81 \text{ m/s}^2)(3.00 \text{ m}) \\ &= 70.6 \text{ J} \checkmark \end{aligned}$$

$$\Delta E_{\text{int}} = \Delta E_{\text{int, block}} + \Delta E_{\text{int, floor}}$$

$$\begin{aligned} \Delta E_{\text{int, floor}} &= \Delta E_{\text{int}} - \Delta E_{\text{int, block}} \checkmark \\ &= 70.6 \text{ J} - 40.0 \text{ J} \\ &= 30.6 \text{ J} \checkmark \end{aligned}$$

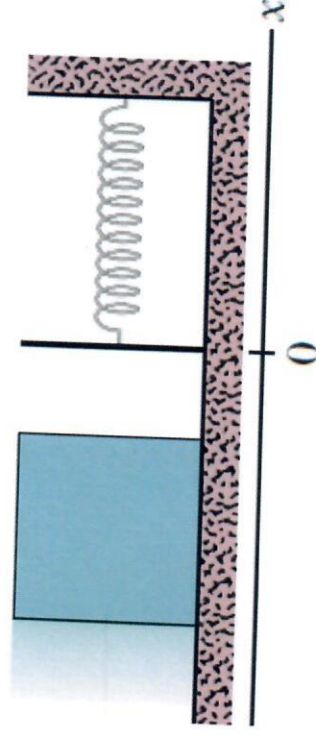
$$\text{c) } W_F = \Delta K + \Delta U + \Delta E_{\text{int}} \checkmark$$

$$\begin{aligned} \Delta K &= W - \Delta U - \Delta E_{\text{int}} \checkmark \\ &= 105 \text{ J} - 0 - 70.6 \text{ J} \checkmark \end{aligned}$$

$$\Delta K = 34.4 \text{ J} \checkmark$$

In Figure, a block of mass $m = 2.5 \text{ kg}$ slides head on into a spring of spring constant $k = 320 \text{ N/m}$. When the block stops, it has compressed the spring by 7.5 cm . The coefficient of kinetic friction between block and floor is 0.25 . While the block is in contact with the spring and being brought to rest:

- what are the work done by the spring force?
- the increase in thermal energy of the block-floor system?
- What is the block's speed just as it reaches the spring?



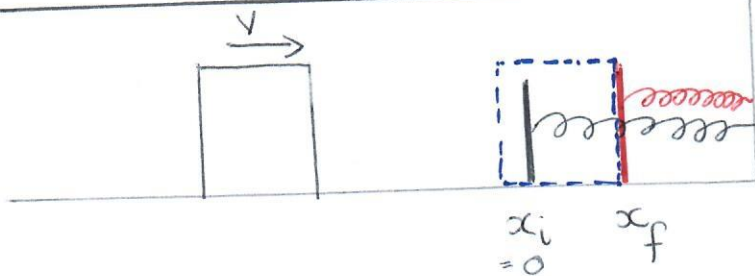
In die figuur, 'n blok met massa 2.5 kg gly na in die rigting van die veer met veerkonstante

$k = 320 \text{ N/m}$. Wanneer die blok tot stilstand kom, het die veer 7.5 cm saamgepers. Die kinetiese-wrywingskoeffisiënt tussen die blok en die vloer is 0.25 . Terwyl die blok in kontak is met die veer en tot rus gebring word:

- wat is die arbeid verrig deur die veer op die blok-vloer stelsel?
- Wat is die toename in termiese energie van die blok-vloer stelsel?
- Wat is die blok se spoed net voor dit kontak maak met die veer?

(10)

Tutorial Test 7-3



$$m = 2.5 \text{ kg}$$

$$k = 320 \text{ N/m}$$

$$\Delta x = 7.5 \text{ cm} = 7.5 \times 10^{-2} \text{ m}$$

$$\mu_k = 0.25$$

Isolated system: block + spring + floor/surface

$$a) W_s = \frac{1}{2} k x_i^2 - \frac{1}{2} k x_f^2 = 0 - \frac{1}{2} (320 \text{ N/m}) (7.5 \times 10^{-2} \text{ m})^2 = -0.90 \text{ J}$$

must be negative

$$b) \Delta E_{\text{int}} = f_k \Delta x = \mu N \Delta x$$

$$\Sigma F_y = m a_y = 0 = N - mg \quad \therefore N = mg$$

$$\therefore \Delta E_{\text{int}} = \mu mg \Delta x = 0.25 (2.5 \text{ kg}) (9.8 \text{ m/s}^2) (7.5 \times 10^{-2} \text{ m}) = 0.46 \text{ J}$$

c) Isolated system:

$$\Delta E_{\text{system}} = 0 = \Delta K + \Delta U + \Delta E_{\text{int}}$$

$$0 = (K_f - K_i) + (U_{gf} - U_{gi}) + (U_{sf} - U_{si}) + \Delta E_{\text{int}}$$

$$W_s = -\Delta U_{sf} \quad K_f = 0, U_g = 0, U_{si} = 0$$

$$\therefore K_i = \frac{1}{2} m v_i^2 = \frac{1}{2} k x_f^2 + \Delta E_{\text{int}}$$

$$v_i = \sqrt{\frac{2 \left[\left(\frac{1}{2} k x_f^2 \right) + \Delta E_{\text{int}} \right]}{m}}$$

$$= \sqrt{\frac{2 [0.90 \text{ J} + 0.46 \text{ J}]}{2.5 \text{ kg}}} = 1.04 \text{ m/s}$$

A spring ($k = 200 \text{ N/m}$) is fixed at the top of a frictionless plane inclined at angle $\theta = 40^\circ$. A 1.0 kg block is projected up the plane, from an initial position that is distance $d = 0.60 \text{ m}$ from the end of the relaxed spring, with an initial kinetic energy of 16 J .

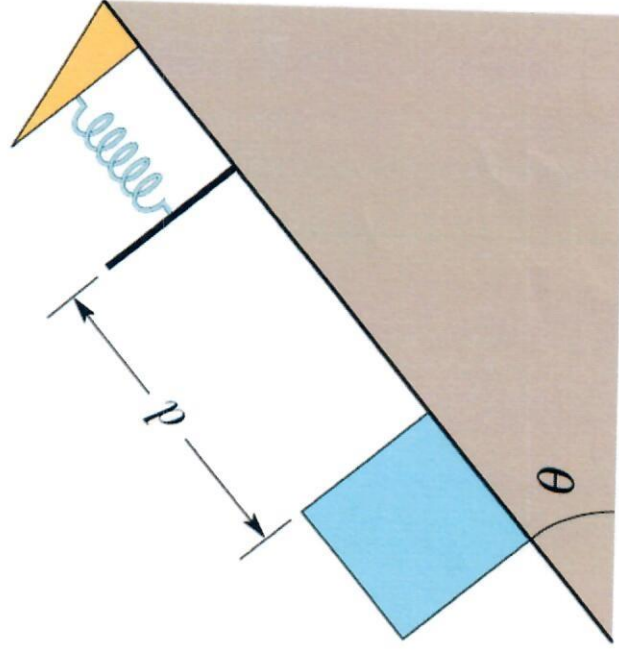
(a) What is the kinetic energy of the block at the instant it has compressed the spring 0.20 m ?

(b) With what kinetic energy must the block be projected up the plane if it is to stop momentarily when it has compressed the spring by 0.40 m ?

'n Veer ($k = 200 \text{ N/m}$) word aan die bopunt van 'n wrywinglose skuinsvlak ($\theta = 40^\circ$) vasgemaak. 'n 1.0 kg blok word by die skuinste geprojekteer, met 'n aanvanklike kinetiese energie van 16 J , van 'n aanvanklike posisie wat $d = 0.60 \text{ m}$ van die eindpunt van die ontspanne veer is.

(a) Wat is die kinetiese energie van die blok die oomblik wanneer dit die veer 0.20 m saamgepers het?

(b) Met watter kinetiese energie moet die blok geprojekteer word by die skuinste op as dit tydelik tot stilstand kom wanneer die veer met 0.40 m saamgepers het? (10)



Tutorial Test 7-4

$$k = 200 \text{ N/m}$$

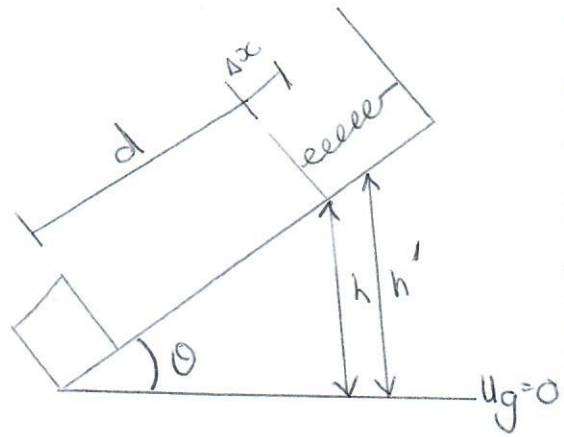
$$\theta = 40^\circ$$

$$m = 1.0 \text{ kg}$$

$$d = 0.60 \text{ m}$$

$$K_i = \frac{1}{2} m v_i^2 = 16 \text{ J}$$

$$\Delta x_s = 0.20 \text{ m}$$



- a) When spring is compressed 0.20 m
 $d' = d + \Delta x = 0.80 \text{ m}$

$$h' = d' \sin \theta = 0.80 \text{ m} \sin 40^\circ$$

$$h' = 0.51 \text{ m} \quad \checkmark$$

Let $U_g = 0$ be at the bottom of the slope.
Isolated system

a) $\Delta E_{\text{mech}} = 0 = \Delta K + \Delta U_g + \Delta U_s = 0$

$$\therefore K_i + U_{gi} + U_{si} = K_f + U_{gf} + U_{sf}$$

$$K_i + 0 + 0 = K_f + mgh' + \frac{1}{2} k x_f^2$$

$$K_f = K_i - mgh' - \frac{1}{2} k x_f^2 \quad \checkmark$$

$$= 16 \text{ J} - 1.0 \text{ kg} (9.8 \text{ m/s}^2) (0.51 \text{ m}) - \frac{1}{2} (200 \text{ N/m}) (0.20 \text{ m})^2$$

$$K_f = 7.0 \text{ J} \checkmark$$

$$b) \Delta E_{\text{mech}} = 0 = \Delta K + \Delta U_g + \Delta U_s$$

$$K_i = K_f + mgh'' + \frac{1}{2} K x^2$$

$$h'' = d'' \sin \theta = (0.6\text{m} + 0.4\text{m}) \sin 40 = 0.64\text{m}$$

$$K_i = 0 + (1.0 \text{ kg})(9.8 \text{ m/s}^2)(0.64 \text{ m}) + \frac{1}{2}(200 \text{ N/m}) \times (0.40 \text{ m})^2$$

$$= 22.3 \text{ J}$$

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